MATH329 · OPERATIONS RESEARCH

Lecture 3 Standard Form

Midterm summary · Chapter 3 · the algebraic solution

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What's inside

01 Why do we need the standard form?

02 The conversion rules

03 The slide examples

04 The midterm question: every rule in one problem

05 Basic solutions: from the graph to algebra

06 Example 3.2: all the basic solutions

07 Example 3.3: with an unrestricted variable

08 Example 3.5: absolute values

09 Common mistakes

10 Test yourself

01 Why do we need the standard form?

The graph works for two variables only. With more variables we solve algebraically with the **Simplex** method, which moves from one corner to a better corner until it reaches the optimum.

> The Simplex Method attempts to move from one corner point of the solution space to a better corner point until the optimum is found.

Before simplex starts, the problem must be written in the **Standard Form** (also called **Equation form**). This is a fixed question in every midterm, worth 5 to 8 marks.

Appeared in the midterm (461)

The question: what does standard form mean? The expected answer, word for word:

> 1. All decision variables are **nonnegative**.
> 2. All constraints (with the exception of the non-negativity of the variables) are **equations with nonnegative RHS**.

02 The conversion rules

Case	What to do	Example
A $\leq$ constraint	Add a **slack variable** with a plus sign.	$x_1 + 3x_2 \leq 6$ $x_1 + 3x_2 + s_1 = 6$
A $\geq$ constraint	Subtract a **surplus variable**, with a minus sign.	$3x_1 + 2x_2 \geq 6$ $3x_1 + 2x_2 - s_2 = 6$
An $=$ constraint	It stays as it is, with no new variable.	$x_1 + x_2 = 5$
A negative right-hand side	Multiply the whole constraint by -1. If it is an inequality, its sign flips.	$3x_1 + 2x_2 \geq -2$ $-3x_1 - 2x_2 \leq 2$
A variable **unrestricted in sign**	Replace it everywhere, including the objective, by the difference of two nonnegative variables.	$x_2 = x_2^+ - x_2^-$
The objective function	Add the new variables with zero coefficients.	$z = 2x_1 + 3x_2 + 0s_1 + 0s_2$
The last line	All variables, old and new, are ≥ 0.	$x_1, x_2, s_1, s_2 \geq 0$

The order matters

First fix the negative right-hand side, then substitute the unrestricted variable, and add the slack and surplus variables last. Adding a slack before multiplying by -1 gives it the wrong sign.

03 The slide examples

1 SLACK Example 3.1

Maximize $z = 2x_1 + 3x_2$
subject to $x_1 + 3x_2 \le 6$
$3x_1 + 2x_2 \le 6$
$x_1, x_2 \ge 0$

Maximize $z = 2x_1 + 3x_2 + 0s_1 + 0s_2$
subject to $x_1 + 3x_2 + s_1 = 6$
$3x_1 + 2x_2 + s_2 = 6$
$x_1, x_2, s_1, s_2 \ge 0$

2 SURPLUS The second constraint is now ≥

Maximize $z = 2x_1 + 3x_2 + 0s_1 + 0s_2$	
subject to $x_1 + 3x_2 + s_1 = 6$	s_1 slack
$3x_1 + 2x_2 - s_2 = 6$	s_2 surplus
$x_1, x_2, s_1, s_2 \ge 0$	

3 NEGATIVE RHS The second constraint is $3x_1 + 2x_2 \geq -2$

$$3x_1 + 2x_2 \geq -2 \quad \times (-1) \quad \rightarrow \quad -3x_1 - 2x_2 \leq 2$$

$$\begin{aligned} &\text{Maximize } z = 2x_1 + 3x_2 + 0s_1 + 0s_2 \\ &\text{subject to} \quad x_1 + 3x_2 + s_1 = 6 \\ &\qquad\qquad -3x_1 - 2x_2 + s_2 = 2 \\ &\qquad\qquad x_1, x_2, s_1, s_2 \geq 0 \end{aligned}$$

4 UNRESTRICTED x_1 is unrestricted

$$\begin{aligned} &\text{Maximize } z = x_1 + 3x_2 \\ &\text{subject to} \quad x_1 + x_2 \leq 2 \\ &\qquad\qquad -x_1 + x_2 \leq 4 \\ &\qquad\qquad x_1 \text{ unrestricted}, x_2 \geq 0 \end{aligned}$$

$$\begin{aligned} &\text{Maximize } z = x_1^+ - x_1^- + 3x_2 + 0s_1 + 0s_2 \\ &\text{subject to} \quad x_1^+ - x_1^- + x_2 + s_1 = 2 \\ &\qquad\qquad -x_1^+ + x_1^- + x_2 + s_2 = 4 \\ &\qquad\qquad x_1^+, x_1^-, x_2, s_1, s_2 \geq 0 \end{aligned}$$

04 The midterm question: every rule in one problem

This question appeared in the same shape in 461 and 462 with different numbers. This is

the 462 version (8 marks):

Maximum $Z = 8x_1 - 2x_2$

S.to $2x_1 + x_2 \leq 12 \cdot x_1 - 5x_2 \geq 6 \cdot 4x_2 = -7$

$x_1 \geq 0$, x_2 unrestricted in sign.

1. What methods can be applied to solve the given linear programming problem?
2. Write the problem in standard form (equation form). "Don't solve the problem"

1 The methods

`Graphical method and Algebraic method.`

The graphical method applies because there are two variables, and the algebraic method always applies.

2 The negative right-hand side

$4x_2 = -7 \quad \times (-1) \quad \rightarrow \quad -4x_2 = 7$

3 Substitute $x_2 = x_2^+ - x_2^-$ everywhere

$-2x_2 \quad \rightarrow \quad -2x_2^+ + 2x_2^-$

$x_2 \quad \rightarrow \quad x_2^+ - x_2^-$

$-5x_2 \quad \rightarrow \quad -5x_2^+ + 5x_2^-$

$-4x_2 \quad \rightarrow \quad -4x_2^+ + 4x_2^-$

4

Add s_1 to the $\leq$ constraint and subtract s_2 from the $\geq$ constraint

Maximum	$Z = 8x_1 - 2x_2^+ + 2x_2^- + 0s_1 + 0s_2$
S.to	$2x_1 + x_2^+ - x_2^- + s_1 = 12$
	$x_1 - 5x_2^+ + 5x_2^- - s_2 = 6$
	$-4x_2^+ + 4x_2^- = 7$
	$x_1, x_2^+, x_2^-, s_1, s_2 \geq 0$

In the answer key: two marks for the methods, two for the objective function, and one for each of the other four lines.

05 Basic solutions: from the graph to algebra

After the standard form we have n variables and m equations, with more variables than equations. To solve:

1. Choose n − m variables and set them equal to zero (nonbasic)
2. Solve the m equations for the remaining m variables (basic)
3. The result is a basic solution
4. If all its values are ≥ 0 it is a Basic Feasible Solution (BFS)

Term	Meaning
Nonbasic variables	The `n − m` variables that we set to zero.
Basic variables	The `m` variables that we solve for.
Basic solution	The unique solution of the system after setting the nonbasic variables to zero.
Basic Feasible Solution	A basic solution with no negative values. Every BFS corresponds to a corner of the feasible region.

The maximum number of basic solutions

$$C(n, m) = \frac{n!}{m!\,(n-m)!}$$

$n = 4, m = 2$: $\quad C(4, 2) = \frac{4!}{2! \cdot 2!} = \frac{24}{4} = 6$

Appeared in the midterm (471)

True/false items, both true:

> A basic feasible solution is a feasible solution to LPP which is also the basic solution. (T)
> A basic feasible solution is called degenerate if value of at least one basic variable is zero. (T)

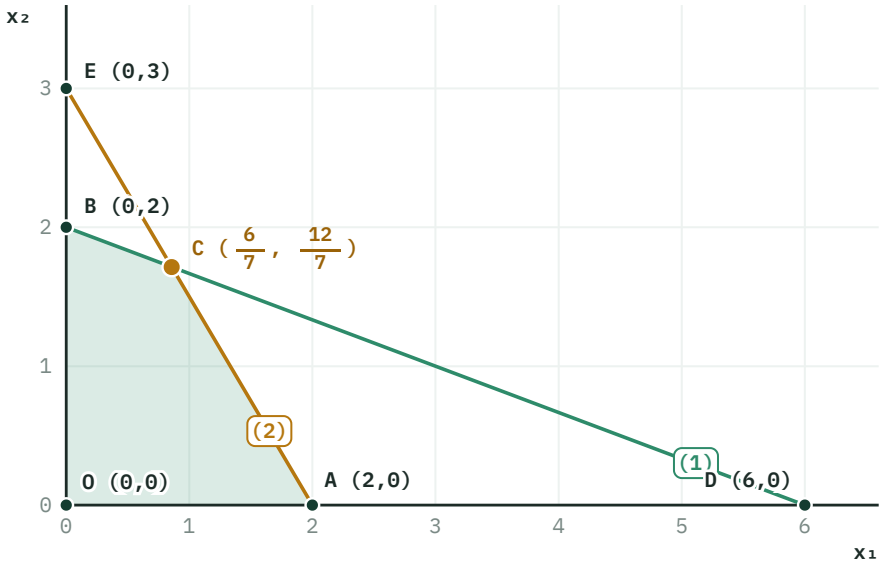
06 Example 3.2: all the basic solutions

Maximize $z = 2x_1 + 3x_2 + 0s_1 + 0s_2$
subject to $x_1 + 3x_2 + s_1 = 6$
$3x_1 + 2x_2 + s_2 = 6$
$x_1, x_2, s_1, s_2 \geq 0$

Here $n = 4$ and $m = 2$, so we set two variables to zero each time and solve for the rest. There are 6 cases:

Nonbasic (= 0)	Basic	Basic solution	Feasible?	z	Corner
s_1, s_2	x_1, x_2	$(\frac{6}{7}, \frac{12}{7}, 0, 0)$	Yes	$\frac{48}{7}$	C
x_2, s_2	x_1, s_1	$(2, 0, 4, 0)$	Yes	4	A
x_2, s_1	x_1, s_2	$(6, 0, 0, -12)$	No	—	D
x_1, s_2	x_2, s_1	$(0, 3, -3, 0)$	No	—	E
x_1, s_1	x_2, s_2	$(0, 2, 0, 2)$	Yes	6	B
x_1, x_2	s_1, s_2	$(0, 0, 6, 6)$	Yes	0	O

The solution is written in the order (x_1, x_2, s_1, s_2). One row as an example: set x_1 and s_1 to zero, so the first equation becomes $3x_2 = 6$ giving $x_2 = 2$, and the second becomes $2(2) + s_2 = 6$ giving $s_2 = 2$.



Optimum $z = \frac{48}{7}$ **at** $x_1 = \frac{6}{7}$, $x_2 = \frac{12}{7}$ **(corner C)**

The two infeasible solutions, D and E, are on the graph but outside the shaded region. The four feasible ones are exactly the corners of the region.

07 Example 3.3: with an unrestricted variable

$$\text{Maximize } z = x_1^+ - x_1^- + 3x_2 + 0s_1 + 0s_2$$

$$\text{subject to} \quad x_1^+ - x_1^- + x_2 + s_1 = 2$$

$$-x_1^+ + x_1^- + x_2 + s_2 = 4$$

$$x_1^+,\ x_1^-,\ x_2,\ s_1,\ s_2 \geq 0$$

Here `n = 5` and `m = 2`, so the maximum is `C(5, 2) = 10`. One case has no solution, so the actual number is 9. The number of basic solutions can be less than the maximum.

Nonbasic (= 0)	Basic	Basic solution	Feasible?	z
x_2, s_1, s_2	x_1^+, x_1^-	no solution	–	–
x_1^-, s_1, s_2	x_1^+, x_2	(−1, 0, 3, 0, 0)	No	–
x_1^-, x_2, s_2	x_1^+, s_1	(−4, 0, 0, 6, 0)	No	–
x_1^-, x_2, s_1	x_1^+, s_2	(2, 0, 0, 0, 6)	Yes	2
x_1^+, s_1, s_2	x_1^-, x_2	(0, 1, 3, 0, 0)	Yes	**8**
x_1^+, x_2, s_2	x_1^-, s_1	(0, 4, 0, 6, 0)	Yes	−4
x_1^+, x_2, s_1	x_1^-, s_2	(0, −2, 0, 0, 6)	No	–
x_1^+, x_1^-, s_2	x_2, s_1	(0, 0, 4, −2, 0)	No	–
x_1^+, x_1^-, s_1	x_2, s_2	(0, 0, 2, 0, 2)	Yes	6
x_1^+, x_1^-, x_2	s_1, s_2	(0, 0, 0, 2, 4)	Yes	0

Optimum $z = 8$ at $x_1^- = 1, x_2 = 3$ so $x_1 = 0 - 1 = -1$, $x_2 = 3$

I computed this table myself, because the slide's table marks one row as feasible although its x_1^- value is negative.

08 Example 3.5: absolute values

Convert the following optimization problem into a LPP:

Maximum $z = \max\{ |2x_1 + 3x_2|, |3x_1 - 7x_2| \}$

Subject to: $x_1 + x_2 \le 2$, $-x_1 + x_2 \le 4$, $x_1, x_2 \ge 0$

The idea: call the larger value y. Since $|a| \le y$ means $-y \le a \le y$, each absolute value gives two linear constraints:

Maximize $Z = y$
subject to $x_1 + x_2 \leq 2$
$-x_1 + x_2 \leq 4$
$2x_1 + 3x_2 - y \leq 0$
$-2x_1 - 3x_2 - y \leq 0$
$3x_1 - 7x_2 - y \leq 0$
$-3x_1 + 7x_2 - y \leq 0$
$x_1, x_2, y \geq 0$

09 Common mistakes

Mistake	Correct
Adding a slack to a $\geq$ constraint	A surplus is subtracted from a $\geq$ constraint: $-s_2$.
Adding a variable to an $=$ constraint	It gets no slack and no surplus.
Leaving a negative right-hand side	$4x_2 = -7$ becomes $-4x_2 = 7$.
Substituting the unrestricted variable in the constraints but not in the objective	It is substituted everywhere it appears.
Losing signs with a negative coefficient	$-5x_2$ becomes $-5x_2^+ + 5x_2^-$.
Forgetting the last line	$x_1, x_2^+, x_2^-, s_1, s_2 \geq 0$

10 Test yourself

1. What does the standard form mean? (two requirements)
2. What is the difference between a slack and a surplus variable?
3. What is the maximum number of basic solutions when $n = 5$ and $m = 3$?
4. Write in standard form: $\text{Min } Z = 3x_1 + 2x_2$ subject to $x_1 + x_2 \leq 9$, $2x_1 - x_2 \geq 4$, $3x_2 = -6$, with $x_1 \geq 0$ and x_2 unrestricted.

Answers

#	Answer
1	All decision variables are nonnegative, and all constraints are equations with nonnegative RHS.
2	A slack is added to a $\leq$ constraint, and a surplus is subtracted from a $\geq$ constraint.
3	$$C(5, 3) = \frac{5!}{3! \cdot 2!} = 10$$
4	$\text{Min } Z = 3x_1 + 2x_2^+ - 2x_2^- + 0s_1 + 0s_2$ $x_1 + x_2^+ - x_2^- + s_1 = 9$ $2x_1 - x_2^+ + x_2^- - s_2 = 4$ $-3x_2^+ + 3x_2^- = 6$ $x_1, x_2^+, x_2^-, s_1, s_2 \geq 0$